

Lecture 12: AI Tools for Scientific Discovery

Experiment design, domain-specific foundation models, and validation

FYS4411 — Reinforcement Learning and Agentic AI
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Scientific discovery is an evidence-producing process

A scientific result is not a fluent explanation. It is a claim connected to observations by a reproducible argument: a model, an experimental or computational protocol, measurements, uncertainty analysis, and tests that could expose error.

Language models can reduce friction in literature work, programming, documentation, and coordination. Agentic systems can connect these capabilities to databases, simulators, instruments, and domain-specific models. Neither capability changes the evidential standard.

The central question is therefore not whether an AI system can propose an experiment. It is whether the complete human–AI workflow produces evidence that is valid, interpretable, and independently checkable.

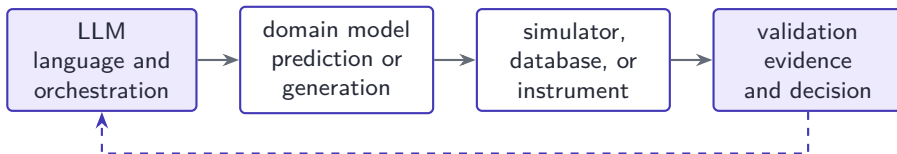
Motivated by Agents for Science curriculum; Messeri and Crockett, *Artificial intelligence and illusions of understanding*.

Learning goals

After this lecture, you should be able to

- ▶ use an LLM or coding agent to formulate, implement, and document a scientific experiment without transferring scientific authority to the model;
- ▶ specify assumptions, controls, observables, uncertainty, validation tests, and stopping conditions before execution;
- ▶ distinguish a general-purpose LLM from a domain-specific foundation model (DSFM) and from an agent built around either model;
- ▶ explain how AlphaFold, Stormer, MatterGen, RETFound, scGPT, Aurora, and Prithvi acquire domain-specific capability;
- ▶ design a validation plan that detects conceptual, numerical, statistical, and operational failure.

The useful system separates four responsibilities



General-purpose models translate intentions, representations, and software tasks. **Domain models** encode regularities in scientific data. **Tools** produce computations or observations. **Validation** determines whether a result may support a claim.

Capability is distributed across the system. Scientific responsibility is not.

Motivated by Compound AI systems; Agents for Science, Lecture 2; Gemini guide supplied with this lecture.

Part	Question
I. Scientific workbench	Which tasks are appropriate for LLMs and coding agents?
II. Experiment design	How can an agent help construct a valid research protocol?
III. Quantum case study	How does the loop work for a driven two-level system?
IV. Domain-specific models	Where does specialized scientific predictive power reside?
V. Applications	How does the architecture transfer across scientific fields?
VI. Validation	What evidence is required before a result is trusted?

The lecture moves from assistance with scientific work to the stronger claim that a model contributes to scientific discovery. The validation burden increases accordingly.

Part I: LLMs as a scientific workbench

Language, code, context, tools, and controlled agency

LLMs are strongest at translation between representations

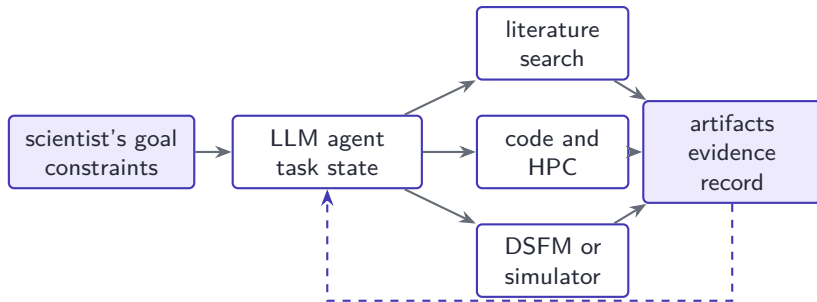
Scientific work repeatedly changes representation:



An LLM can propose mappings between these forms, explain APIs, draft local functions, summarize a method, or turn a protocol into a checklist. These mappings are useful because they reduce coordination cost.

The scientific risk is that a syntactically smooth mapping can preserve vocabulary while changing the model: a sign, unit, boundary condition, estimator, or causal assumption may be altered without producing a software error.

A scientific workbench is a compound system



The LLM does not need to contain the relevant measurement, simulation, or prediction internally. It needs a typed interface to the component that does, plus control logic that preserves provenance and constraints.

Motivated by BAIR, *The Shift from Models to Compound AI Systems*; Agents for Science curriculum.

Different research tasks require different safeguards

Task	Useful assistance	Required safeguard
Literature mapping	queries, clustering, evidence table	inspect primary sources; record search coverage
Hypothesis generation	alternatives and mechanisms	novelty search; falsifiability; expert screening
Protocol design	factors, controls, measurements	feasibility, power, safety, preregistered decisions
Programming	local functions, tests, documentation	review diffs; analytical and regression tests
Data analysis	scripts, diagnostics, visualizations	frozen data; independent implementation; uncertainty
Interpretation	competing explanations	distinguish observation, inference, and speculation
Writing	structure and technical editing	verify every claim, number, citation, and attribution

The closer a task is to an evidential or operational decision, the less appropriate it is to rely on linguistic plausibility alone.

Chat, IDE assistance, and agent mode are distinct

	Chat	IDE assistant	Agent mode
Primary value	planning and review	local implementation	multi-step repository tasks
Context	prompt and attachments	files, selections, terminal	workspace plus tools
Typical action	proposes text	proposes edits	edits, runs, and inspects
Main risk	unsupported reasoning	plausible wrong code	unintended state change
Control	ask assumptions	inspect every diff	permissions, budgets, version control

The attached Gemini guide recommends moving from conceptual planning in chat to small, local, testable edits in the IDE. Agent mode should be used only when the objective and admissible changes are narrow.

Motivated by *Using Gemini for Coding in Scientific Work*, supplied course guide.

Context is part of the computational experiment

For a model call

$$y \sim p_{\theta}(y \mid c),$$

the context c includes instructions, selected files, previous messages, retrieved papers, terminal output, and tool results. A change in context can change generated code or scientific interpretation even when the nominal task is unchanged.

Reproducible use therefore records

- ▶ model and interface version, date, prompt, and relevant parameters;
- ▶ files or data exposed to the model and explicit exclusions;
- ▶ tool calls, returned artifacts, accepted edits, and human interventions;
- ▶ the tests or observations used to accept the result.

Private data, credentials, unpublished sensitive results, and irrelevant raw data should not enter the context.

Motivated by Gemini guide: workspace context, context exclusion, and AI use log.

A scientific prompt should function as a contract

Minimal prompt schema

Objective + scientific assumptions + allowed methods + input/output contract + validation evidence + scope and stopping rule.

For example:

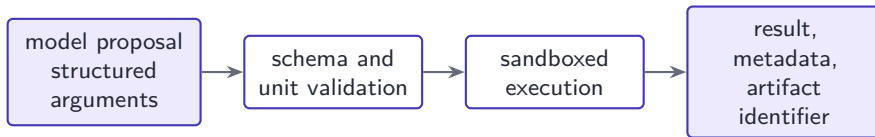
Implement the smallest readable function for $H = (\Delta/2)\sigma_z + (\Omega/2)\sigma_x$. Use NumPy only, $\hbar = 1$, complex arrays, and shape (2,2). Do not add time dependence. Then propose tests for Hermiticity and the zero-drive limit; do not interpret physical results.

The prompt exposes decisions that would otherwise be filled by defaults. It does not guarantee correctness; it makes deviations easier to detect.

Tool calls require typed scientific interfaces

A tool action should be a structured operation, not an instruction hidden in prose:

```
 $a_t = \text{simulate}(\text{model\_id}, \theta, \text{solver}, \text{tolerances}, \text{seed}).$ 
```



Validation before execution checks types, units, ranges, permissions, and resource budgets. Validation after execution checks status, convergence, completeness, provenance, and scientific invariants.

Agent mode changes the authority boundary

An ordinary completion proposes text. An agent may read files, modify code, invoke a terminal, submit a job, query a database, or operate an instrument. The relevant risk is therefore the product of error probability and action consequence.

$$\text{risk}(a) \approx P(\text{incorrect } a) C(\text{consequence of } a).$$

- ▶ Use least privilege: grant only the files, tools, and commands required.
- ▶ Require approval for irreversible, costly, safety-critical, or externally visible actions.
- ▶ Run under version control, preserve logs, and make changes recoverable.
- ▶ Define a budget and a termination condition before the agent begins.

Broad requests such as “improve this repository” are scientifically under-specified and operationally over-authorized.

Motivated by Gemini guide: agent mode; agent-system safety principles.

AI assistance should leave a provenance trail

A useful record distinguishes proposal, acceptance, execution, and verification:



An `AI_USE.md` or laboratory log should record substantial generated implementations, refactors, analyses, or protocol changes and state how each was checked. The model is not a scientific source and should not be represented as a co-author.

Transparency makes later debugging possible: it reveals where unverified assumptions entered the workflow.

Motivated by Gemini guide: declaring and documenting use.

Part II: Designing a research experiment

From a question to an executable and falsifiable protocol

Experiment design is constrained decision-making

Let d denote a design, y a possible observation, and ϑ uncertain scientific parameters. A useful design maximizes expected information or utility under cost, safety, and feasibility constraints:

$$d^* = \arg \max_{d \in \mathcal{D}} \mathbb{E}_{y \sim p(y|d)} [U(d, y)] \quad \text{s.t.} \quad C(d) \leq B, R(d) \leq R_{\max}.$$

An LLM can help enumerate candidate designs and make assumptions explicit. A simulator, statistical model, or domain foundation model can estimate outcomes. The scientist defines the utility, admissible evidence, and constraints.

An experiment is not improved by generating more protocol text unless the generated design changes the expected information, feasibility, or assurance.

Four objects must not be conflated

Object	Example
Research question	Does pulse-shape robustness improve population transfer under detuning?
Hypothesis	A smooth bounded pulse produces higher mean fidelity over the specified detuning distribution.
Measurement	Final $ 1\rangle$ population, leakage, duration, and calibration uncertainty.
Decision rule	Prefer design A only if the preregistered confidence interval for improvement excludes zero and constraints are satisfied.

LLM-generated plans often blur these categories. The result is a protocol that collects data without specifying what observation would support, weaken, or reject the hypothesis.

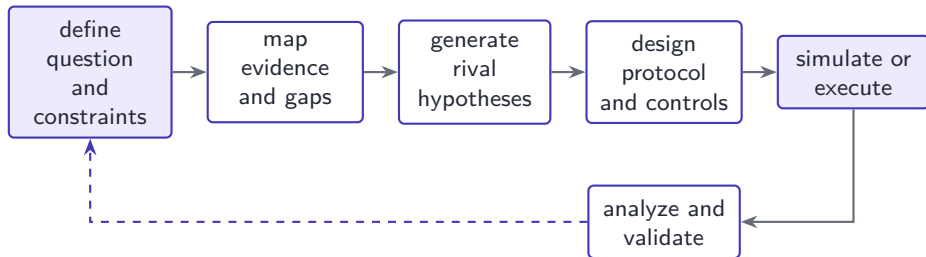
A design card makes the protocol inspectable

Before code or instrument control, write a one-page design card:

- ▶ **Question and mechanism:** what causal or predictive relation is being tested?
- ▶ **Model and scope:** equations, approximations, population, regime, and exclusions.
- ▶ **Factors and controls:** manipulated variables, baselines, randomization, and blinding.
- ▶ **Observables:** units, acquisition, preprocessing, and uncertainty.
- ▶ **Analysis:** estimator, comparison, multiplicity, and decision rule.
- ▶ **Validity checks:** positive controls, negative controls, analytical limits, and replication.
- ▶ **Operations:** budget, safety, approval gates, and stopping conditions.

The LLM may critique this card, but changes should be explicit and versioned.

An agent can support the complete design loop



At each transition, the system should produce an artifact: an evidence table, hypothesis set, protocol version, executable configuration, raw result, or validation report. Conversational history alone is not an adequate laboratory record.

Literature tools should produce an evidence matrix

Source	Population / system	Intervention	Outcome	Limitations
Paper A	hardware family and regime	pulse family	fidelity and leakage	single device; no drift
Paper B	simulated open system	robust control	expected fidelity	assumed noise model
Paper C	independent experiment	calibrated baseline	measured error rate	different readout pipeline

The evidence matrix forces retrieval to preserve identifiers, methods, and limitations. It also exposes missing comparisons and conflicting definitions.

A narrative summary is useful only after the underlying claims have been traced to primary sources. Citation presence alone does not establish that the cited source supports the claim.

Hypothesis generation should preserve alternatives

For an observation O , ask the system to construct a set of mechanisms

$$\mathcal{H} = \{H_1, H_2, \dots, H_k\}$$

such that each H_i implies at least one discriminating prediction. A useful output is not a ranked list of plausible stories but a contrastive table:

Hypothesis	Predicted signature	Discriminating measurement
H_1 : detuning error	phase-dependent loss	scan detuning at fixed amplitude
H_2 : amplitude drift	time-correlated bias	interleaved calibration controls
H_3 : leakage	population outside subspace	leakage-sensitive readout

The agent should retain unresolved disagreement rather than collapse early onto the most fluent mechanism.

Measurements should be chosen for discrimination

One formal criterion is expected information gain:

$$\text{EIG}(d) = \mathbb{E}_{y \sim p(y|d)} \left[D_{\text{KL}}(p(\vartheta | y, d) \parallel p(\vartheta)) \right].$$

An agent can use a simulator or surrogate to estimate which design d best separates competing hypotheses. The result depends on the prior, likelihood, candidate design space, and utility definition; these are scientific inputs, not generic language-model outputs.

Practical questions include:

- ▶ Does the measurement respond differently under the rival mechanisms?
- ▶ Can the instrument resolve that difference at the available sample size?
- ▶ Does the design confound the effect with time, batch, operator, or preprocessing?

Generate code in small, scientifically named components

The Gemini guide recommends delaying code until the workflow is checked, then implementing one local function at a time.

Weak request

Simulate the experiment and make the plots.

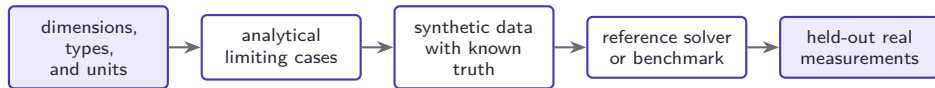
Constrained request

Implement only the constant-Hamiltonian propagator. Preserve complex dtype, validate shapes, and use $U = \exp(-iHT)$. Do not add plotting or optimization.

Small components reduce the semantic surface of each review. They make it possible to connect a function directly to an equation and to tests with known answers.

Motivated by Gemini guide, Sections 4–5 and 10.

Validation should proceed from exact to realistic



Each level detects different errors. Passing a realistic-looking plot is not a substitute for passing a conservation law or analytical limit. Agreement with a reference implementation is not independent validation if both share the same data leakage or convention.

Validation evidence should be specified before the model generates the implementation whenever possible.

A computational experiment is a versioned artifact set

```
experiment/  
  src/model.py  
  src/analysis.py  
  tests/test_model.py  
  configs/baseline.yaml  
  data/README.txt  
  run_experiment.py  
  AI_USE.md  
  environment.lock  
  results/manifest.json
```

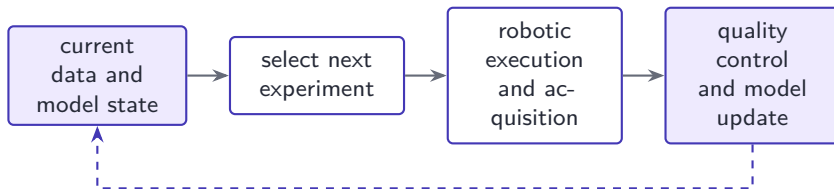
The agent should modify this structure through narrow, reviewable changes and return a manifest of affected files and commands.

The repository separates

- ▶ model implementation from analysis;
- ▶ executable parameters from source code;
- ▶ tests from generated results;
- ▶ immutable raw data from derived artifacts;
- ▶ AI assistance from scientific attribution.

Motivated by Adapted from the project structure and AI use log in the Gemini guide.

Self-driving laboratories close the physical loop



In chemistry and materials science, an autonomous loop may select a composition, synthesize it, characterize the product, and update a surrogate model. The value comes from sequential experimental design, not from removing the scientist from the system.

Physical execution requires calibration checks, failure recovery, safety interlocks, sample tracking, and explicit bounds on admissible actions.

Motivated by Agents for Science, Lecture 6; *Self-Driving Laboratories for Chemistry and Materials Science*.

Human approval belongs at consequential transitions

Transition	Approval question	Evidence presented
Question → protocol	Is this the intended scientific claim?	design card and alternatives
Protocol → execution	Is it feasible, safe, and sufficiently controlled?	parameters, controls, budget
Analysis → interpretation	Do results pass preregistered checks?	raw data, diagnostics, uncertainty
Interpretation → publication	Are claims supported and reproducible?	provenance and independent review

Approval should not be a generic “looks good” click. It should answer a defined scientific or operational question using a compact evidence bundle.

Wrong and right requests differ in evidential structure

Wrong use	Better use
“Find a novel hypothesis and prove it.”	Generate rival hypotheses, cite primary evidence, and state discriminating experiments.
“Write the complete simulation.”	State assumptions, then implement one equation-linked function and tests.
“Optimize the experiment.”	Define objective, constraints, uncertainty distribution, and a baseline first.
“Interpret this significant result.”	Inspect preprocessing, effect size, uncertainty, multiplicity, and alternatives.
“Run whatever tools are needed.”	Enumerate allowed tools, budgets, approval gates, and stopping conditions.

The better requests do not merely contain more detail. They assign different epistemic roles to proposal, execution, and validation.

Part III: Quantum experiment case study

Designing and validating a robust driven-qubit protocol

The question is robust population transfer

Consider a driven two-level system initialized in $|0\rangle$. The nominal task is to choose a bounded pulse that transfers population to $|1\rangle$. The scientific question is stronger:

Research question

Which pulse parameters maintain high final-state fidelity under a specified distribution of detuning and amplitude errors, without exceeding duration and amplitude constraints?

The experiment must therefore include a nominal baseline, an uncertainty model, robustness measurements, and a decision rule. Maximizing one simulated trajectory is not sufficient.

This example follows the quantum workflow in the attached Gemini guide, extended from a simple Rabi sweep to constrained robust design.

The model fixes the convention

Assume $\hbar = 1$ and a rotating-frame Hamiltonian

$$H(t) = \frac{\Delta}{2}\sigma_z + \frac{\Omega(t)}{2}\sigma_x, \quad i\frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

For a constant pulse, the propagator is

$$U(T) = e^{-iHT}, \quad |\psi(T)\rangle = U(T)|0\rangle.$$

The factor $1/2$, the use of angular frequency, the basis ordering, and the sign convention are part of the model contract. A generated implementation that changes any of these may still produce smooth oscillatory plots.

The observables connect the model to a decision

The target-state probability and pure-state fidelity are

$$P_1(T) = |\langle 1 | \psi(T) \rangle|^2, \quad F(T) = |\langle \psi_{\text{target}} | \psi(T) \rangle|^2.$$

For hardware, the observed frequency is estimated from counts,

$$\hat{P}_1 = \frac{n_1}{N}, \quad \text{SE}(\hat{P}_1) \approx \sqrt{\frac{\hat{P}_1(1 - \hat{P}_1)}{N}},$$

before additional corrections for readout error, drift, and repeated calibration.

The protocol should state whether the decision concerns ideal fidelity, measured population, leakage-corrected fidelity, or a composite operational metric.

Factors, controls, and responses define the experiment

Role	Variable	Planned treatment
Control factor	pulse amplitude and duration	grid, optimizer, or parameterized pulse
Nuisance factor	detuning Δ	sampled from specified range or distribution
Nuisance factor	amplitude scale error ϵ	replace $\Omega \mapsto (1 + \epsilon)\Omega$
Baseline	resonant square π pulse	$\Delta = 0, \Omega T = \pi$
Negative control	zero duration or zero drive	identity or no-transfer limit
Response	P_1 , fidelity, leakage, cost	report nominal and distributional summaries

The LLM can draft this table. The scientist must justify the uncertainty ranges and their relationship to the intended experimental regime.

The LLM and numerical tools have different roles

LLM or coding agent

- ▶ formulate file and function structure;
- ▶ translate equations into local code;
- ▶ generate candidate tests;
- ▶ diagnose selected errors;
- ▶ document assumptions and commands.

Scientific tools

- ▶ matrix exponential or ODE solver;
- ▶ optimizer and uncertainty sampler;
- ▶ simulator or quantum backend;
- ▶ statistical analysis;
- ▶ independent reference calculation.

The model proposes how to use the tools; the tools execute defined operations; tests and measurements determine whether the proposal is acceptable.

Generated code should remain equation-linked

```
def hamiltonian(delta: float, omega: float) -> np.ndarray:
    """H = (delta/2) sigma_z + (omega/2) sigma_x; hbar = 1."""
    return 0.5 * delta * SIGMA_Z + 0.5 * omega * SIGMA_X

def evolve_constant(psi0, H, duration):
    if psi0.shape != (2,) or H.shape != (2, 2):
        raise ValueError("incompatible two-level shapes")
    if duration < 0:
        raise ValueError("duration must be non-negative")
    return expm(-1j * H * duration) @ psi0
```

The docstring records the convention. Shape checks expose representation errors. The implementation remains small enough to compare line-by-line with $U(T) = \exp(-iHT)$.

Code generation is useful here because the expected behavior is precise and independently testable.

Motivated by Adapted from the driven-qubit implementation in the Gemini guide.

Analytical tests are stronger than plausible output

```
def test_zero_duration_is_identity():
    H = hamiltonian(delta=0.3, omega=1.2)
    assert np.allclose(evolve_constant(KET_0, H, 0.0), KET_0)

def test_norm_is_preserved():
    H = hamiltonian(delta=0.4, omega=1.0)
    psi = evolve_constant(KET_0, H, 2.0)
    assert np.isclose(np.linalg.norm(psi), 1.0)

def test_resonant_pi_pulse():
    H = hamiltonian(delta=0.0, omega=1.0)
    psi = evolve_constant(KET_0, H, np.pi)
    assert np.isclose(abs(np.vdot(KET_1, psi))**2, 1.0)
```

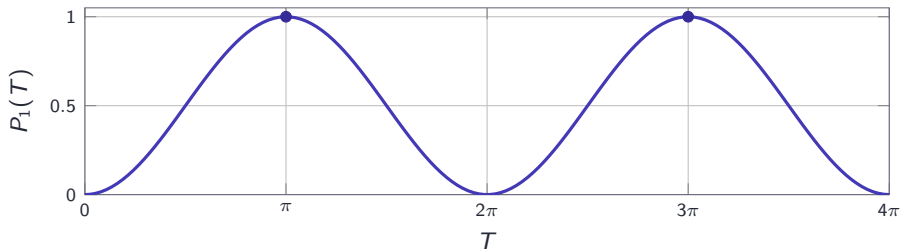
The last test is valid only for the stated Hamiltonian convention. Tests encode scientific assumptions and must change when those assumptions change.

Motivated by Gemini guide, quantum-system workflow.

The resonant solution provides a visual invariant

For $\Delta = 0$ and constant $\Omega = 1$,

$$P_1(T) = \sin^2(T/2).$$



Maxima at odd multiples of π and norm preservation are inexpensive checks that should precede any optimizer or agent loop.

A pulse sweep is a designed computational experiment

For each configuration, save parameters and outcomes rather than only a figure:

$$\mathcal{D} = \{(T_i, \Delta_j, \epsilon_k, P_1, F, \text{status}, \text{seed})\}.$$

- ▶ Choose a duration grid that resolves the expected oscillation period.
- ▶ Include the analytical resonant slice and off-resonant slices.
- ▶ Separate optimizer-search data from final held-out evaluation points.
- ▶ Record solver tolerances and convergence status.
- ▶ Save a machine-readable table before generating plots.

The agent may write the sweep script and plotting code, but the experimental design determines what the script must sample and preserve.

Robust design requires an explicit objective

Let ϕ parameterize a pulse and $\xi = (\Delta, \epsilon)$ denote uncertainty. One objective is

$$J(\phi) = \mathbb{E}_{\xi \sim p(\xi)}[1 - F(\phi; \xi)] + \lambda_T T(\phi) + \lambda_A \int_0^T |\Omega_\phi(t)|^2 dt.$$

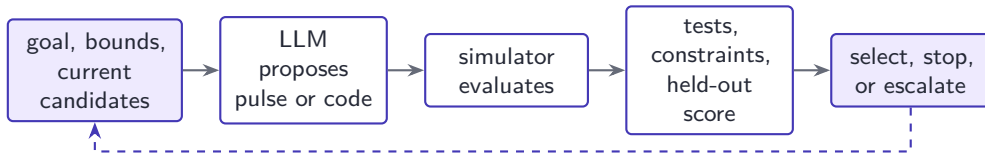
Possible alternatives include worst-case loss,

$$J_{\max}(\phi) = \max_{\xi \in \Xi} [1 - F(\phi; \xi)],$$

or a chance constraint $P(F \geq F_{\min}) \geq 1 - \alpha$.

An agent cannot infer the scientifically appropriate risk functional from “make the pulse robust.” Mean, worst-case, and tail guarantees answer different questions.

The agent loop searches, but verifiers decide



The verifier should reject non-Hermitian models, constraint violations, solver failures, data leakage, and candidates that improve only the search set. The final design is evaluated on a frozen uncertainty set and compared with the baseline.

This is the same structural idea used in code-search systems such as FunSearch and AlphaEvolve: generative proposals become useful when coupled to executable evaluators.

A factor-of-two error can look scientifically plausible

With the stated convention,

$$H = \frac{\Omega}{2} \sigma_x \implies P_1(T) = \sin^2\left(\frac{\Omega T}{2}\right),$$

so the first maximum occurs at $T = \pi/\Omega$.

If generated code uses $H = \Omega \sigma_x$, it predicts

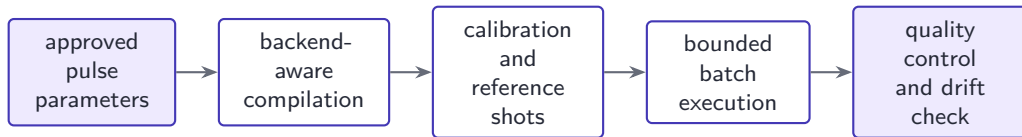
$$P_1^{\text{wrong}}(T) = \sin^2(\Omega T),$$

with a first maximum at $T = \pi/(2\Omega)$. Both curves are smooth, normalized, and periodic. A visual check alone may accept the wrong model.

Correct validation

Tie the implementation to the equation, derive one analytical reference point, and test it before exploring new regimes.

Hardware execution adds calibration and safety state



The agent should not autonomously reinterpret calibration failures or increase shot budgets without a policy. Instrument limits, queue cost, credentials, and externally visible jobs require explicit permissions.

Simulation validates software logic. Hardware data validates whether the model and calibration describe the physical device in the relevant regime.

A complete result includes its reconstruction record

For each reported figure or table, preserve

- ▶ the model equation and convention version;
- ▶ code commit, environment lock, configuration, seed, and solver metadata;
- ▶ raw counts or raw simulation outputs and immutable identifiers;
- ▶ preprocessing and analysis commands;
- ▶ failed runs, exclusions, and reasons;
- ▶ prompts or agent actions that materially changed the protocol or code;
- ▶ the acceptance tests and human approvals.

The final plot is a view of an artifact graph. Reproducibility requires the graph, not merely the image.

Part IV: Domain-specific foundation models

Scientific representations, transferable predictors, and specialist agents

A DSFM is pretrained on a scientific domain

A **domain-specific foundation model** is trained on broad data from a scientific domain and reused across multiple downstream tasks through prompting, conditioning, fine-tuning, adapters, or task heads.

The pretraining data may consist of protein sequences and structures, atmospheric fields, crystal graphs, satellite image time series, retinal images, gene-expression profiles, spectra, trajectories, or simulation states. Text may be present, but it is not the defining modality.

The term “foundation” refers to transfer across tasks within a domain, not to universal competence. A large task-specific predictor with no credible reuse may be valuable, but it is not automatically a foundation model.

Motivated by Agents for Science, Lecture 2; scientific foundation-model literature.

Scientific tokens encode domain objects and symmetries

Domain	Model input	Relevant structure
Proteins	amino acids, MSA, templates	sequence, geometry, evolutionary covariation
Weather	multilevel spatial fields	sphere, pressure levels, coupled variables, time
Materials	atoms, coordinates, lattice	periodicity, permutation, rotation, composition
Single cells	ranked gene expression	sparse counts, cell state, perturbation, batch
Earth observation	multispectral image sequences	spatial scale, season, sensor, geography
Robotics	images, language, actions	embodiment, temporal control, contact dynamics

Domain-specific representations and inductive biases determine which regularities can be learned efficiently. Treating every object as ordinary text may discard geometry, units, conservation, uncertainty, or measurement structure.

Pretraining learns a reusable scientific representation

Let x denote an unlabeled domain object and m a corruption or masking operator. A generic self-supervised objective is

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{x,m} [\mathcal{L}(f_{\theta}(m(x)), x)] .$$

For a downstream task with limited labels (x_i, y_i) , adapt a small task component g_{ϕ} :

$$\phi^* = \arg \min_{\phi} \sum_i \ell(g_{\phi}(f_{\theta^*}(x_i)), y_i).$$



The scientific value is label efficiency and transfer, provided the pretraining distribution contains structure relevant to the downstream regime.

A DSFM becomes an agent only inside a control loop

AlphaFold, Stormer, or MatterGen is a model, not an agent by itself. It becomes a **specialist agent** when a system gives it

- ▶ a defined role and task state;
- ▶ tools for acquiring inputs or executing follow-up calculations;
- ▶ an action space, such as selecting a candidate, requesting a measurement, or proposing a design;
- ▶ feedback from verifiers, experiments, or downstream models;
- ▶ a budget and stopping or escalation condition.

Formally, the DSFM may provide a domain policy or scoring function

$$a_t \sim \pi_{\text{domain}}(a \mid s_t),$$

while the orchestrator owns state transitions, permissions, and acceptance.

The strongest architecture is often hybrid

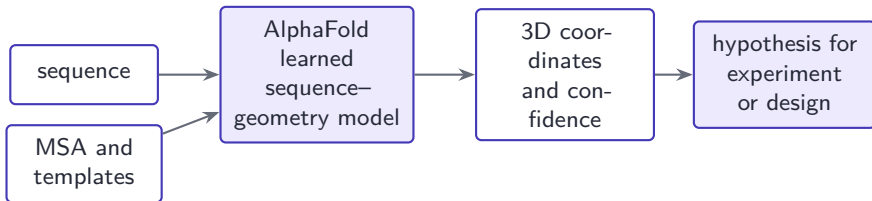


The LLM handles open-ended language, decomposition, code, and interfaces. The DSFM handles a bounded scientific mapping learned from domain data. Conventional software enforces schemas and policies. Experiments or trusted solvers arbitrate scientific claims.

This decomposition makes each component replaceable and separately evaluable.

AlphaFold maps biological sequence to structure

AlphaFold2 predicts three-dimensional protein structure using the amino-acid sequence, multiple-sequence alignments, and optional templates. Its architecture combines learned evolutionary and geometric representations and directly predicts atomic coordinates.



The relevant scientific capability is not linguistic knowledge about proteins. It is a learned mapping constrained by sequence, evolutionary information, and three-dimensional geometry.

Motivated by Jumper et al., *Highly accurate protein structure prediction with AlphaFold*, Nature (2021).

AlphaFold output is a prediction

Confidence measures such as pLDDT and predicted aligned error help identify uncertain local structure and relative domain placement. They do not establish that

- ▶ the predicted conformation is the relevant state under the experimental conditions;
- ▶ a mutation, ligand, post-translational modification, or complex is represented correctly;
- ▶ dynamics, disorder, binding affinity, or mechanism follows from the static prediction;
- ▶ a result is suitable for clinical use.

Right use

Use confidence-aware predictions to prioritize hypotheses, design constructs, interpret complementary data, or select experiments; validate consequential claims with structural, biochemical, or functional evidence.

Motivated by AlphaFold publication and official implementation guidance.

Stormer learns atmospheric dynamics across lead times

Stormer is a transformer for medium-range global weather forecasting. Its domain-specific design includes weather-variable embeddings, a pressure-weighted loss, and a randomized dynamics objective over forecast intervals.

For state x_t and sampled interval δ , training approximates

$$\hat{x}_{t+\delta} = f_{\theta}(x_t, \delta), \quad \mathcal{L}(\theta) = \sum_{v,p} w_p \ell(\hat{x}_{v,p}, x_{v,p}),$$

where v indexes atmospheric variables and p pressure levels.

At inference, different sequences of forecast intervals can be combined for a target lead time. The model's tokens are multivariate atmospheric patches, not words.

Motivated by Nguyen et al., *Scaling transformer neural networks for skillful and reliable medium-range weather forecasting*, NeurIPS (2024).

Weather models require regime-aware validation

Check	Scientific question
Hindcast split	Were evaluation years and reanalysis products held out correctly?
Operational baseline	Does the model improve on persistence, climatology, and numerical weather prediction?
Spatial and variable metrics	Is skill consistent across regions, levels, variables, and lead times?
Extremes	Are tail events, cyclone tracks, and rare regimes represented?
Calibration	Do predicted probabilities or ensembles match observed frequencies?
Physical diagnostics	Are mass, energy, spectra, and coupled fields plausible over rollout?

Average global error can improve while local extremes or out-of-distribution regimes degrade. Operational value depends on the decision and region, not on one aggregate score.

Earth foundation models transfer across tasks

Aurora

Pretrained on diverse geophysical data, then adapted to weather, air-pollution, ocean-wave, and cyclone-related forecasting tasks. The foundation-model claim rests on transfer across Earth-system prediction tasks.

In an agent, these models can propose high-value locations or time windows for further observation. An acquisition policy must still account for cloud cover, sensor constraints, geographic shift, and downstream decision cost.

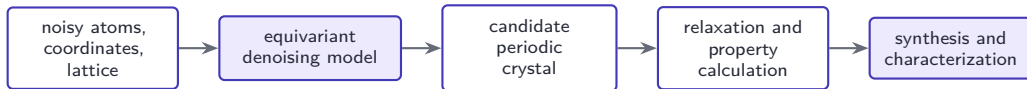
Prithvi-EO

Pretrained on multispectral satellite image time series and adapted to downstream Earth-observation tasks such as segmentation and change analysis. Spatial, spectral, temporal, and sensor structure replace natural-language tokens.

Motivated by Microsoft Research, *Aurora: A Foundation Model for the Earth System*; NASA–IBM Prithvi-EO-2.0.

MatterGen generates periodic crystal structures

MatterGen is a diffusion model over atom types, coordinates, and lattice variables. It can be fine-tuned or conditioned toward property targets, enabling inverse materials design.



Generation changes the search distribution; it does not establish stability, synthesizability, novelty, or useful measured properties.

Motivated by Zeni et al., *A generative model for inorganic materials design*, Nature (2025).

Materials discovery needs a cascade of filters

For candidate c , a staged decision may be

$$c \xrightarrow{\text{DSFM}} \hat{p}(c) \xrightarrow{\text{DFT/MD}} p_{\text{sim}}(c) \xrightarrow{\text{synthesis}} \text{sample} \xrightarrow{\text{assay}} p_{\text{meas}}(c).$$

- ▶ Fast learned filters remove obviously poor or duplicate candidates.
- ▶ Higher-fidelity simulation checks relaxation, stability, and target properties.
- ▶ Synthesis planning checks precursors, conditions, hazards, and feasibility.
- ▶ Characterization measures phase purity and the actual property of interest.
- ▶ Negative results update the surrogate and the acquisition policy.

The agent's purpose is to allocate expensive validation efficiently, not to replace it with a model score.

Biomedical DSFMs learn from biological measurements

Model	Pretraining domain	Representative transfer
scGPT	gene-expression profiles from many cells	cell-type annotation, integration, perturbation-related tasks
RETFound	large collections of unlabeled retinal images	label-efficient adaptation to ocular and systemic disease tasks
AlphaFold	sequences, alignments, structures	structure prediction across protein targets

These models learn statistical structure from biological data rather than clinical or mechanistic truth. External validation must address cohort, instrument, ancestry, disease prevalence, acquisition protocol, and intended use.

A specialist biomedical agent may query a DSFM, retrieve supporting evidence, and propose an assay; it must not silently convert a research prediction into a clinical decision.

Motivated by Cui et al., scGPT, Nature Methods (2024); Zhou et al., RETFound, Nature (2023).

DSFMs differ in what they transfer

Model	Domain object	Transfer mechanism	Critical validation
AlphaFold	sequence and structure	shared structural predictor	confidence, complexes, experimental state
Stormer	atmospheric state	rollout across lead times	hindcasts, extremes, calibration
Aurora	Earth-system fields	fine-tuning across tasks	task and regional generalization
MatterGen	periodic crystals	conditional generation	relaxation, novelty, synthesis, assay
RETFound	retinal images	task-specific adaptation	external cohorts and clinical utility
scGPT	single-cell profiles	embeddings and task heads	batch, perturbation, biological validity
Prithvi-EO	satellite time series	fine-tuning for EO tasks	sensor, geography, season, resolution

There is no domain-independent validation protocol. Each model’s scientific object and downstream decision determine the relevant failure modes.

Domain models carry the scientifically relevant compression

A general LLM compresses regularities in language and code. A DSFM compresses regularities in a scientific observation space. This difference matters because the highest-value scientific tasks often require

- ▶ prediction from high-dimensional, non-text measurements;
- ▶ generation under geometric, chemical, or physical constraints;
- ▶ transfer where labeled data are scarce or experiments are expensive;
- ▶ fast surrogates for repeated simulation or design;
- ▶ uncertainty estimates tied to a domain and a decision.

LLMs remain important as interfaces and orchestrators. The domain model is often where the system obtains predictive competence that directly changes which experiment is performed.

Specialization does not remove epistemic limits

A DSFM may fail because

- ▶ training data omit the relevant regime or inherit systematic measurement bias;
- ▶ the representation discards variables required for the downstream mechanism;
- ▶ a proxy objective is optimized instead of the scientific quantity of interest;
- ▶ uncertainty is miscalibrated under distribution shift;
- ▶ generated objects are valid in silico but infeasible in the laboratory;
- ▶ a high average score hides important subgroups, spatial regions, or rare events.

Operational rule

Use a DSFM to prioritize scarce attention and expensive experiments. Do not treat its output as the terminal evidence unless the downstream claim is itself about model behavior.

Part V: Scientific applications

The same architecture, different evidence and failure modes

Medicine requires prospective and external validation



An LLM may draft inclusion criteria, code data checks, and generate a reporting checklist. A DSFM may supply image or omics representations. Neither substitutes for endpoint validity, external cohorts, subgroup analysis, calibration, prospective testing, or clinical governance.

Wrong use: infer diagnosis directly from a research model output. **Right use:** evaluate a prespecified decision-support role against current practice and measure patient-relevant outcomes.

Chemistry couples generation to synthesis and assay

For molecular or materials discovery, the useful loop is

generate → filter → simulate → synthesize → measure → update.

An LLM agent can translate the target into constraints, retrieve protocols, write simulation inputs, and maintain the experiment state. A chemistry or materials model can generate structures or predict properties. A robotic platform can execute bounded procedures.

Validation must distinguish

- ▶ chemical validity from novelty;
- ▶ computed stability from synthesizability;
- ▶ target-property prediction from measured performance;
- ▶ one successful sample from a reproducible process window.

Energy systems combine forecasts with decisions

Weather and Earth-system models can inform renewable generation, load, hazard, and maintenance decisions. The scientific system is a coupled forecast–optimization problem:

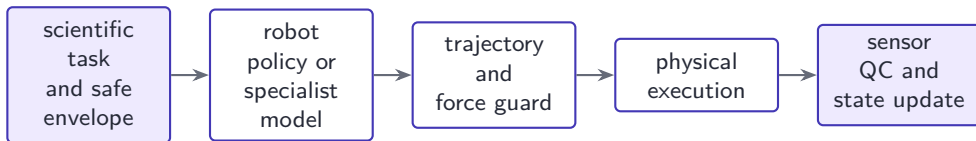
$$u^* = \arg \min_u \mathbb{E}_{w \sim p_{\text{DSFM}}(w|x)} [C(u, w)] \quad \text{s.t. grid and reliability constraints.}$$

The forecast model estimates scenarios w ; the optimizer chooses a decision u ; conventional constraints enforce feasibility. An LLM may orchestrate data retrieval, scenario analysis, diagnostics, and reporting.

Forecast accuracy should be evaluated through the downstream decision as well as meteorological error. Calibration and extreme-event performance can matter more than small improvements in mean error.

Robotics turns model error into physical consequence

Robotic foundation models learn reusable representations or policies from diverse trajectories, images, language, and actions. Open X-Embodiment, for example, aggregates more than one million real-robot trajectories across many embodiments.



Simulation, low-speed trials, force and workspace limits, emergency stops, and human approval are part of the model evaluation.

Motivated by Open X-Embodiment and RT-X project.

Biology benefits from model-guided experimental prioritization

A hybrid biological discovery system might

1. use an LLM agent to map literature and define a mechanism-level question;
2. query AlphaFold or a sequence/structure model for structural hypotheses;
3. query a single-cell model for candidate cell states or perturbations;
4. rank candidates under assay cost, uncertainty, and novelty constraints;
5. execute binding, expression, perturbation, or functional assays;
6. update the hypothesis using positive and negative evidence.

The DSFM reduces the candidate space. The assay determines whether the relevant biological effect occurs. Mechanistic interpretation may require additional perturbation and replication.

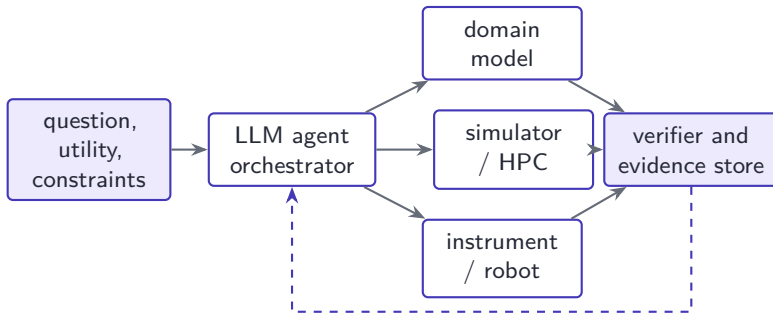
Quantum science is ready for tool-rich agents

Quantum workflows already expose structured tools: circuit simulators, tensor-network codes, electronic-structure packages, pulse optimizers, error-correction simulators, cloud backends, and hardware calibration data.

Agent role	Bounded scientific task
Planner	select solver and approximation under resource limits
Coder	implement observable, ansatz, decoder, or analysis with tests
Experimenter	submit approved parameter sweeps and preserve job metadata
Critic	compare symmetries, limits, convergence, and independent methods
Coordinator	allocate HPC and hardware runs under a fixed budget

Domain foundation models for quantum systems are less standardized than AlphaFold or weather models. Near-term value therefore comes from integrating mature scientific software with explicit verification.

The cross-domain architecture is stable



What changes across medicine, chemistry, energy, robotics, biology, and quantum science is the domain object, tool contract, risk, and acceptable evidence. The control pattern remains recognizable.

Choose the simplest component that answers the question

Need	Appropriate component
Explain, structure, translate, or write local code	general-purpose LLM
Retrieve a known paper, datum, or protocol	database or search tool
Compute a defined numerical quantity	deterministic scientific software
Predict or generate in a specialized data domain	validated DSFM
Select sequential experiments under uncertainty	acquisition model plus agent loop
Execute physical actions	instrument runtime with safety interlocks
Accept or reject a scientific claim	prespecified analysis and human/domain review

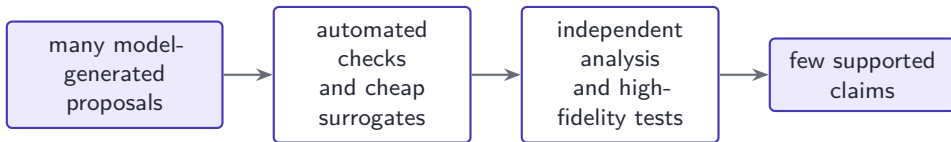
Adding an agent or foundation model is justified only when it improves measured success, cost, information gain, or assurance relative to a simpler baseline.

Part VI: Validation is part of the method

From plausible output to defensible scientific evidence

AI increases proposal throughput faster than validation

Agents can generate literature maps, hypotheses, code, candidate molecules, protocols, and analyses at low marginal cost. Experimental replication, expert review, high-fidelity simulation, and instrument time remain expensive.



The resulting bottleneck is not ideation but triage and validation. Scientific infrastructure must allocate verification effort according to consequence, novelty, uncertainty, and expected value.

Motivated by Google DeepMind, *AI agents and the new validation bottleneck in science*; Messeri and Crockett (2024).

Wrong and right use differ by the acceptance rule

Setting	Wrong use	Right use
LLM literature review	accept a coherent uncited synthesis	trace claims to primary sources and preserve search coverage
Generated code	run once and trust a plausible plot	review equation-linked components and analytical tests
AlphaFold	treat predicted structure as measured mechanism	use confidence-aware structure to prioritize experiments
Stormer / weather FM	cite one global mean score	evaluate lead time, region, variable, extremes, and calibration
MatterGen	report generated crystals as discoveries	relax, deduplicate, assess synthesis, and measure properties
Medical DSFM	deploy from retrospective performance	external, subgroup, prospective, and utility validation
Agent mode	allow broad autonomous changes	bound tools, budgets, approvals, logging, and recovery

The same output can be responsible or irresponsible depending on what decision it is allowed to trigger.

Uncertainty must be tested under distribution shift

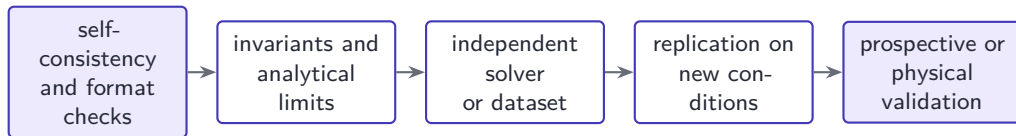
For prediction $\hat{y}$ with reported uncertainty $q_\alpha(x)$, calibration on one distribution does not imply calibration on a shifted target distribution:

$$P_{(x,y) \sim P_{\text{train}}}(y \in q_\alpha(x)) \approx 1 - \alpha \not\Rightarrow P_{(x,y) \sim P_{\text{target}}}(y \in q_\alpha(x)) \approx 1 - \alpha.$$

Shift may arise from a new instrument, site, patient population, weather regime, material family, laboratory protocol, simulator version, or hardware calibration.

Validation should stratify performance by the variables that define intended use and include explicit abstention or escalation when inputs fall outside the supported domain.

Evidence should be independent of the proposal mechanism



Confidence increases when evidence relies on different assumptions, data, code, instruments, or investigators. Asking the same model to critique its own unsupported claim is useful screening, but it is weak independence.

The necessary level depends on consequence. A classroom helper function and a clinical or safety-critical claim require different evidence.

Reproducibility requires an artifact graph

Represent a scientific run as a directed graph

$$G = (V, E), \quad V = \{\text{data, code, configuration, model, prompt, result, claim}\}.$$

An edge records a transformation with version, parameters, time, environment, and responsible actor.

- ▶ Use immutable identifiers for raw data and generated artifacts.
- ▶ Record model and DSFM checkpoints, not only product names.
- ▶ Preserve negative results, retries, exclusions, and manual corrections.
- ▶ Make every published figure reconstructible from a declared command.
- ▶ Record substantial AI use and the validation attached to it.

A transcript is useful provenance, but it is not a substitute for executable configurations and data lineage.

Scientific agents require data and action governance

Risk	Failure	Control
Sensitive data	context exposes private or unpublished information	exclusion, access control, minimization, audit
Prompt injection	retrieved text is interpreted as an instruction	instruction–data separation and content labeling
Unsafe action	agent chooses harmful parameters or procedures	semantic constraints, interlocks, human approval
Resource misuse	loops consume HPC, instrument time, or reagents	budgets, quotas, timeouts, stopping rules
Silent mutation	code, data, or protocol changes without trace	version control, immutable raw data, manifests
Model misuse	research output is used outside validated scope	use statements, gating, abstention, monitoring

Security and scientific validity meet at the action boundary: an invalid action can corrupt both the experiment and its evidence record.

Evaluation must include lab-style and field-style tasks

The EAIRA methodology distinguishes complementary evaluation classes:

1. **multiple-choice evaluation** for factual knowledge and basic concepts;
2. **open-response evaluation** for technical reasoning and problem solving;
3. **lab-style experiments** for controlled, instrumented research-assistant tasks;
4. **field-style experiments** for real researcher interactions across domains.

A scientific agent should additionally be evaluated on trajectory validity, tool-use correctness, cost, reproducibility, recovery from failure, and whether human users detect and correct erroneous suggestions.

Benchmark success is evidence about a benchmark. Deployment claims require evaluation in the intended scientific workflow.

Motivated by Cappello et al., *EAIRA: Establishing a Methodology for Evaluating AI Models as Scientific Research Assistants* (2025).

The objective is accelerated, not automated, understanding

1. Use LLMs for language, code, decomposition, and orchestration.
2. Use domain-specific models where prediction or generation depends on scientific data and structure.
3. Convert every important output into an inspectable artifact with provenance.
4. Define controls, analytical anchors, uncertainty, acceptance, and stopping before execution.
5. Allocate human review and independent validation according to consequence.
6. Treat negative results and detected failures as information for the next design.

Central conclusion

AI tools matter scientifically when they increase the rate at which reliable evidence is produced. More proposals, code, or predictions are useful only when the validation system can distinguish discovery from plausible error.

Selected sources and links

Theme	Representative source
Curriculum	Agents for Science: https://agents4science.github.io/Class/curriculum.html
Scientific coding	<i>Using Gemini for Coding in Scientific Work</i> , supplied course guide
AI and understanding	Messori and Crockett (2024): https://doi.org/10.1038/s41586-024-07146-0
AlphaFold	Jumper et al. (2021): https://doi.org/10.1038/s41586-021-03819-2
Stormer	Nguyen et al. (2024): https://arxiv.org/abs/2312.03876
Aurora	https://www.microsoft.com/en-us/research/publication/aurora-a-foundation-model-for-the-earth-system/
MatterGen	Zeni et al. (2025): https://doi.org/10.1038/s41586-025-08628-5
Biomedical DSFMs	RETFound: https://doi.org/10.1038/s41586-023-06555-x ; scGPT: https://doi.org/10.1038/s41592-024-02201-0
Earth observation	Prithvi-EO-2.0: https://github.com/NASA-IMPACT/Prithvi-EO-2.0
Robotics	Open X-Embodiment: https://robotics-transformer-x.github.io/
Scientific agents	Co-Scientist: https://arxiv.org/abs/2502.18864 ; AlphaEvolve: https://deepmind.google/blog/alphaevolve-a-gemini-powered-coding-agent-for-designing-advanced-algorithms/
Evaluation	EAIRA: https://arxiv.org/abs/2502.20309

Additional motivating material: BAIR compound AI systems; Agents for Science lectures on self-driving laboratories, human–AI workflows, and evaluation.